Name: _____ Period: _____

WHY WE STUDY EARTH & SPACE

Directions: Your group has a decision to make. There's just one problem… You don't have all the information. Your job isn't to find the "right" answer. Your job is to make the best decision you can with the evidence available. Read your group's scenario carefully and discuss the situation together. Make a decision based on the information you have, then complete the questions below as a group. Be prepared to share your decision and explain your reasoning to the class. Remember, there may not be one correct answer.

THE DECISION: What would your group do?

_____

_____

OUR REASONING: List three reasons your group made this decision. Use information from the scenario to explain the thinking behind your choice.

1. _____

_____

2. _____

_____

3. _____

_____

WHAT ELSE DO WE NEED TO KNOW?: List 3 pieces of information that would help you make a better decision.

1. _____

2. _____

3. _____

WHAT WOULD CHANGE YOUR MIND? Describe one piece of new information or evidence that could cause your group to make a different decision.

_____

_____

_____

Name: _____ Period: _____

EXIT TICKET: WHY WE STUDY EARTH & SPACE

1. Before today, I thought Earth Science was mostly about:

_____

_____

2. Now, one reason Earth Science matters is...

_____

_____

because...

_____

_____

3. Based on today's challenges, why is it important to study Earth & Space Science?

_____

_____

Name: _____ Period: _____

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SCENARIO 1



Your family finds a house for sale for $100,000 less than similar houses nearby. It looks great, has plenty of space, and is only a short walk from a river. The realtor says the owners are motivated to sell and that several other families are interested.

The house has never had water inside it, according to the current owners. However, your family notices that the backyard slopes toward the river, and a neighbor mentions that the river gets very high during some spring storms.

Your family loves the house, but the realtor says you need to make an offer by tomorrow.

Your decision: Would you recommend buying the house?

Consider:

- What information would you want before making an offer?
- What could explain the unusually low price?
- Does the fact that the house has never flooded mean it is safe from future flooding?
- What evidence would make you change your decision?

SCENARIO 2



It is 8:00 PM. The forecast says there is a 70% chance of heavy snow overnight, but only a few inches are currently on the ground. Some forecasts predict 4 inches by morning while others predict more than 18 inches.

You are the superintendent and must decide whether to close school tomorrow. Some students live only a few minutes from school, while others travel on rural roads or take buses from farther away. The forecast also says the heaviest snow may not begin until early morning.

Closing school unnecessarily creates problems for families, but waiting too long could mean buses are already on the road when conditions become dangerous.

Your decision: What do you do right now?

Consider:

- Is a 70% chance of snow enough information to make your decision?
- What additional information matters besides snowfall totals?
- Why might different locations experience different conditions?
- When would you need to make the final decision?
- What evidence would make you change your decision?

SCENARIO 3



Your family finds an incredible deal on a vacation near a tropical beach. Three days before you leave, meteorologists begin tracking a tropical storm hundreds of miles away. The storm might strengthen and might move toward your destination, but forecasters aren't certain.

Canceling now costs $500. Waiting could mean losing the entire cost of the trip. The airline is still operating normally, and the hotel says it is too early to know whether the storm will affect the area.

One forecast shows the storm passing far from your destination. Another shows it moving much closer. Your family has to decide whether to cancel now, wait for more information, or continue with the trip.

Your decision: Cancel, wait, or go?

Consider:

- What information matters most before deciding?
- How certain would scientists need to be before you acted?
- How do the possible risks and costs affect your decision?
- What would you continue monitoring?
- What evidence would make you change your decision?

SCENARIO 4



Your family is driving through the mountains. It has been raining heavily for two days. Your GPS gives you two choices:

- Route A: 45 minutes, winding mountain road
- Route B: 2 hours, major highway

Both roads are currently open. Route A is much faster, but it passes along steep hillsides and through several narrow valleys. Route B is longer and has more traffic, but it is a major highway.

Your family wants to get to the hotel before dark. Your GPS does not show any accidents or road closures, but you notice water running down the hillside beside the road.

Your decision: Which route do you take?

Consider:

- Is “currently open” enough evidence that a road is safe?
- What would you want to know about the landscape?
- Why might the previous two days of rain matter?
- What information could your GPS be missing?
- What evidence would make you change your decision?

SCENARIO 5 

A town gets most of its drinking water from underground. A company wants to build a new factory outside town and promises 200 new jobs. Many residents support the project because the town has lost several businesses in recent years.

The factory would store chemicals on the property. The company says everything will be stored according to safety regulations and that the facility will be inspected regularly.

The proposed site is several miles from the town's drinking-water wells. Some residents argue that this distance makes the factory safe, while others worry that a spill could eventually affect the water supply.

The town board must vote on whether to approve the project.

Your decision: Should the town approve the factory?

Consider:

- What information would you want before voting?
- Does the location of the factory matter?
- Could a spill affect the town's water even if the factory is several miles away?
- How would you balance the economic benefits against possible environmental risks?
- What evidence would make you change your decision?

SCENARIO 6 

A growing town needs a new hospital. Two locations are available.

- Site A: Flat, inexpensive land near a river
- Site B: Higher ground farther from town that will cost $8 million more to develop

The hospital is expected to remain there for at least 75 years. Site A is closer to most residents and would allow the hospital to open sooner. Site B is farther away, but it is located on higher ground.

The town has experienced flooding in the past, although neither site has been seriously damaged. Some residents argue that the extra $8 million should be spent on medical equipment instead of land.

Your decision: Which site would you choose?

Consider:

- Should today's conditions be the only consideration?
- What natural events could affect either site?
- How does the expected 75-year lifetime of the hospital affect your decision?
- What information about the past could help you predict future risk?
- What evidence would make you change your decision?

SCENARIO 7



A volcano near a city hasn't erupted in hundreds of years. Recently, scientists have detected small earthquakes underneath it. The volcano looks exactly the same as it did last month.

Scientists say the earthquakes could be a sign of activity beneath the volcano, but they cannot yet say whether an eruption will occur. Some residents want to leave immediately, while others argue that the volcano has been quiet for generations.

Evacuating the nearby area would affect thousands of people and cost millions of dollars. Waiting could avoid an unnecessary evacuation, but it could also leave less time to respond if conditions change.

Your decision: Would you evacuate?

Consider:

- Does “nothing has happened yet” mean nothing will happen?
- What evidence would you want scientists to collect?
- What are the consequences of evacuating unnecessarily?
- What are the consequences of waiting too long?
- What evidence would make you change your decision?

SCENARIO 8



Scientists discover an asteroid that will pass near Earth in 15 years. Early calculations show a small possibility that it could collide with Earth, but scientists need more observations before they can determine its exact path.

Preparing a mission to change its path would cost billions of dollars. Some scientists argue that planning should begin now because a mission could take years to design and launch. Others argue that the money should not be spent until the risk is better understood.

New observations over the next several years may show that the asteroid will miss Earth completely, or they may show that the risk is greater than originally thought.

Your decision: When, if ever, should we begin preparing?

Consider:

- How much certainty should we require before spending billions of dollars?
- Does having 15 years change your decision?
- What additional observations would help scientists understand the risk?
- What are the consequences of waiting for more information?
- What evidence would make you change your decision?